AOTMS AI CALLING

COMPLETE PRODUCT REQUIREMENTS DOCUMENT (PRD)

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TABLE OF CONTENTS

1. Product Overview
2. Product Vision and Objectives
3. Target Users and User Roles
4. Application Navigation and Modules
5. Dashboard
6. Employees
7. New Employee Creation
8. Employee Management Workspace
9. Talk to an Employee
10. Instant Leads
11. Bulk Campaigns
12. Inbound Calls
13. Leads & Results
14. All Conversations
15. Train Employees
16. Phone Numbers
17. Performance
18. Billing
19. Settings

20. Developers
21. Integrations
22. Technical Architecture
23. Technology Stack
24. Programming Architecture
25. Complete Folder Structure
26. Database Architecture
27. API Architecture
28. AI Voice Processing Architecture
29. Workflow Automation
30. Security Requirements
31. Non-Functional Requirements
32. Testing Requirements
33. Deployment and Infrastructure
34. Development Roadmap
35. Development Time Estimation
36. MVP Scope
37. Acceptance Criteria
38. Reverse Engineering Validation
39. Future Scope

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1. PRODUCT OVERVIEW

1.1 Introduction

OWN AI is a multilingual AI Calling SaaS platform designed to help businesses automate inbound and outbound voice conversations through intelligent AI employees.

The platform enables organizations to create, configure, train, deploy, and manage AI voice employees capable of performing business operations such as:

- Lead Generation
- Customer Support

- Sales Calling
- Appointment Scheduling
- Follow-up Calls
- Customer Qualification
- Information Collection
- Business Communication
- Inbound Call Handling
- Outbound Call Campaigns

Users can create AI employees using custom prompts, select supported languages, configure voices, select AI models, provide knowledge sources, and connect phone numbers.

The platform will provide a centralized dashboard to monitor calls, leads, conversations, employee performance, usage, and billing.

1.2 Product Vision

To build a scalable multilingual AI Calling platform that enables businesses across India and other markets to automate voice-based customer interactions using configurable AI employees.

1.3 Product Goals

1. Build a complete AI Employee creation and management system.
2. Support multilingual AI voice conversations, targeting 15+ Indian languages.
3. Enable inbound and outbound AI calling.
4. Provide instant lead calling and bulk campaign management.
5. Allow users to train AI employees with business-specific knowledge.
6. Provide call analytics, transcripts, recordings, and summaries.
7. Support CRM and third-party integrations.
8. Automate business workflows using n8n.
9. Provide organization-level management and multi-tenant security.
10. Implement subscription and usage-based billing.

1.4 Product Design Principles

- Simple and intuitive user experience.
- Modular and maintainable architecture.
- Secure multi-tenant design.

- Provider-independent AI integration.
- Scalable calling infrastructure.
- Reliable background processing.
- API-first backend.
- Clear usage and billing transparency.

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2. TARGET USERS

The platform is intended for:

- Small and Medium Businesses.
- Sales and Marketing Teams.
- Customer Support Teams.
- Call Centers.
- Educational Institutions.
- Hospitals and Clinics.
- Insurance and Finance Companies.
- Real Estate Businesses.
- Travel and Tourism Businesses.
- Local Service Providers.
- Jewellery and Fashion Businesses.
- Enterprise Organizations.

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3. USER ROLES AND PERMISSIONS

3.1 Super Admin

Platform-level administrator.

Responsibilities:

- Manage organizations.
- Manage platform users.
- Manage subscription plans.
- Monitor platform usage.

- Manage AI providers.
- Monitor calling infrastructure.
- Manage platform configurations.
- Access platform-level analytics.

3.2 Organization Owner

Responsibilities:

- Manage organization.
- Manage subscription and billing.
- Create and manage employees.
- Manage team members.
- Configure integrations.
- View organization analytics.

3.3 Organization Admin

Responsibilities:

- Manage organization operations.
- Create AI employees.
- Manage campaigns.
- Manage leads.
- Manage calling configuration.
- Manage team permissions according to assigned access.

3.4 Manager

Responsibilities:

- Monitor AI employee performance.
- Manage leads.
- Review conversations.
- Monitor campaigns.
- Manage follow-ups.
- View operational reports.

3.5 Team Member

Responsibilities depend on permissions:

- Access assigned employees.
- View permitted leads.
- Review assigned conversations.
- Perform permitted calling operations.

All roles must be protected by Role-Based Access Control (RBAC).

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4. APPLICATION NAVIGATION AND MODULES

The application will use the following navigation structure.

OVERVIEW

1. Dashboard
2. Employees
3. New Employee
4. Talk to an Employee

CALLING

1. Instant Leads
2. Bulk Campaigns
3. Inbound Calls

RESULTS & SETUP

1. Leads & Results
2. All Conversations
3. Train Employees
4. Phone Numbers
5. Performance
6. Billing
7. Settings
8. Developers

---

5. MODULE 01 – DASHBOARD

5.1 Purpose

The Dashboard provides a centralized overview of business operations, AI employees, calling activity, leads, conversations, and performance.

5.2 Dashboard Components

A. Quick Actions

Provide shortcuts to frequently used actions.

- Create New Employee.
- Start Instant Lead Calling.
- Create Bulk Campaign.
- Configure Inbound Calls.
- Upload Leads.
- View Conversations.

B. Your AI Employees

Display configured AI employees.

Required fields:

- Employee Name.
- Avatar.
- Status.
- Supported Languages.
- Assigned Voice.
- AI Model.
- Total Calls.
- Success Rate.
- Last Active Time.

Actions:

- View Employee.
- Edit Employee.
- Test Employee.
- Activate / Deactivate.

C. Needs Your Attention

Display alerts and pending operational actions.

Examples:

- Failed calls.
- Campaign failures.
- Incomplete employee configuration.
- Low account credits.
- Phone number issues.
- Integration failures.
- Pending follow-ups.

D. Recent Summary

Display recent business activities:

- Total Calls.
- Leads Generated.
- Successful Conversations.
- Appointments Scheduled.
- Pending Follow-ups.
- Human Transfers.

E. At A Glance

Display important KPIs:

- Total Calls.
- Answered Calls.
- Missed Calls.
- Successful Calls.
- Total Leads.
- Qualified Leads.
- Average Call Duration.
- Credits Used.

F. Trends

Display interactive analytical charts.

Requirements:

- Daily view.
- Weekly view.
- Monthly view.
- Custom date range.
- Interactive tooltips.
- Data filters.

G. Calls Over Time

Display calling activity over time.

Metrics:

- Total Calls.
- Inbound Calls.
- Outbound Calls.
- Answered Calls.
- Failed Calls.

Visualization: Line Chart.

H. Busiest Hours

Display the busiest calling periods.

Visualization: Bar Chart.

I. How Conversations Ended

Display conversation outcomes.

Categories:

- Successful.
- Lead Captured.
- Appointment Booked.
- Interested.
- Not Interested.
- No Answer.

- Busy.
- Failed.
- Transferred to Human.
- Other.

Visualization: Donut / Pie Chart.

J. Where Conversations Came From

Display conversation sources.

Sources:

- Instant Leads.
- Bulk Campaigns.
- Inbound Calls.
- Google Sheets.
- Meta Lead Ads.
- Manual Upload.
- API Integration.
- Other Integrations.

Visualization: Donut Chart.

K. Manager Insights

Display operational insights.

Examples:

- Best-performing AI employees.
- Highest-converting campaigns.
- Calling hours with better response rates.
- Employees requiring additional training.
- Pending follow-ups.
- Failed campaign alerts.

AI-generated insights are a proposed enhancement.

L. Recent Calls

Display recent calls in a table.

Required columns:

Field	Description
Contact	Customer name and number
Employee	Assigned AI employee
Channel	Inbound / Outbound
Outcome	Final call result
Sentiment	Customer sentiment
Captured	Captured information
Talk Time	Conversation duration
Credits	Credits consumed
Speed to Lead	Time to first call attempt
Time	Call date and time
Summary	Conversation summary

Actions:

- View Conversation.
- View Transcript.
- Listen to Recording.
- View Contact.
- Export Data.

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6. MODULE 02 – EMPLOYEES

6.1 Purpose

The Employees module manages all AI employees created by an organization.

Each employee represents an independently configurable AI voice agent.

6.2 Employee Listing

Display:

- Employee Name.

- Employee Avatar.
- Description.
- Languages.
- Voice.
- AI Model.
- Status.
- Total Calls.
- Success Rate.
- Creation Date.

Actions:

- View.
- Edit.
- Duplicate.
- Activate.
- Deactivate.
- Delete.

Filters:

- Employee Name.
- Status.
- Language.
- Creation Date.

---

7. MODULE 03 – NEW EMPLOYEE CREATION

7.1 Purpose

Allow users to create custom AI employees through a guided configuration workflow.

7.2 Employee Identity

Fields:

- Employee Name.
- Employee Description.

- Avatar.
- Employee Purpose.
- Business Category.

7.3 Language Selection

Requirements:

- Select one or multiple languages.
- Support configured Indian languages.
- Validate language availability.
- Validate voice compatibility.
- Configure default conversation language.
- Support language-specific prompts.

The initial target is 15+ Indian languages, subject to provider capabilities and testing.

7.4 Custom Prompt Editor

Users must be able to create custom AI instructions in any language supported by the selected AI model.

Requirements:

- System Prompt Editor.
- Employee Personality.
- Employee Role.
- Business Instructions.
- Conversation Objectives.
- Greeting Instructions.
- Customer Interaction Rules.
- Response Style.
- Conversation Closing Instructions.
- Prompt Editing.
- Prompt Version History.

7.5 Voice Commands – Speech to Text

Allow users to provide employee instructions using voice input.

Requirements:

- Microphone permission.
- Audio recording.
- Speech-to-text conversion.
- Editable transcription.
- Insert transcription into prompt editor.

7.6 AI Model Selection

Allow users to select an available AI model.

Display:

- Model Name.
- Provider.
- Language Support.
- Model Capabilities.
- Configuration Options.

Model availability must be validated through configured provider integrations.

7.7 Built-in Employee Templates

Provide predefined business templates.

Initial categories:

1. Insurance & Finance.
2. Travel & Tourism.
3. Hospital & Clinic.
4. Coaching & Tuitions.
5. Real Estate.
6. Local Services.
7. Jewellery & Fashion.

Each template should contain:

- Template Name.
- Description.
- Suggested Employee Role.
- Default Prompt.

- Suggested Languages.
- Suggested Voice.
- Conversation Guidelines.
- Suggested Outcomes.

Users must be able to customize templates before saving.

7.8 Build Employee

Requirements:

- Validate required fields.
- Validate language and voice compatibility.
- Validate AI model selection.
- Save employee configuration.
- Display creation status.
- Navigate to employee workspace after successful creation.

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8. MODULE 04 – EMPLOYEE MANAGEMENT WORKSPACE

After creating an employee, users can open a dedicated employee workspace.

The workspace will contain:

1. Overview.
2. Instant Leads.
3. Call Script.
4. Training.
5. Actions.
6. Outcomes.
7. Voices.
8. Settings.

8.1 Employee Overview

Display:

- Employee Identity.
- Status.

- Languages.
- Voice.
- AI Model.
- Total Calls.
- Successful Calls.
- Average Call Duration.
- Recent Conversations.

Actions:

- Edit Employee.
- Test Employee.
- Activate / Deactivate.

8.2 Instant Leads

Allow users to initiate individual calls using the selected employee.

Fields:

- Customer Name.
- Phone Number.
- Lead Information.
- Selected Language.
- Employee.

Actions:

- Start Calling.
- View Calling Status.
- View Result.
- Schedule Follow-up.

8.3 Call Script

Manage employee conversation instructions.

Requirements:

- Edit System Prompt.
- Configure Greeting.

- Define Conversation Objectives.
- Define Questions.
- Define Conversation Stages.
- Configure Objection Handling.
- Configure Closing Instructions.
- Preview Script.
- Test Script.
- Save Versions.

8.4 Training

Allow users to provide employee-specific knowledge.

Proposed requirements:

- PDF Upload.
- DOCX Upload.
- TXT Upload.
- Website URL.
- FAQ Management.
- Knowledge Source Management.
- Processing Status.
- Remove Source.
- Test Knowledge Responses.

Knowledge retrieval may use Retrieval-Augmented Generation (RAG).

8.5 Actions

Configure actions that an AI employee can execute.

Proposed actions:

- Capture Lead Information.
- Schedule Appointment.
- Create Follow-up.
- Send Notification.
- Transfer to Human.

- Trigger Webhook.
- Update CRM.
- Send WhatsApp Message.

Actions must have validation, permissions, and execution logs.

8.6 Outcomes

Configure conversation outcomes.

Examples:

- Interested.
- Not Interested.
- Qualified Lead.
- Unqualified Lead.
- Appointment Scheduled.
- Callback Requested.
- Follow-up Required.
- No Answer.
- Failed.
- Human Transfer.

Requirements:

- Create Custom Outcomes.
- Edit Outcomes.
- Configure Descriptions.
- Map AI-detected Outcomes.
- Store Outcomes with Call Records.
- Use Outcomes in Analytics.

8.7 Voices

Configure AI employee voices.

Requirements:

- Provider Selection.
- Voice Selection.

- Language Compatibility.
- Voice Preview.
- Voice Configuration.
- Speaking Style Settings, if supported.

8.8 Settings

Manage employee-specific settings.

- Employee Name.
- Description.
- Language.
- Voice.
- AI Model.
- Prompt.
- Calling Preferences.
- Employee Status.
- Delete Employee.

---

9. MODULE 05 – TALK TO AN EMPLOYEE

9.1 Purpose

Allow users to test AI employees before production deployment.

Requirements:

- Select Employee.
- Start Test Conversation.
- Test Voice Responses.
- Test Language.
- Test Prompt Instructions.
- Test Knowledge Base.
- View Transcript.
- Review Conversation Behavior.

Test calls must be clearly identified and separated from production billing unless explicitly configured otherwise.

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10. MODULE 06 – INSTANT LEADS

10.1 Purpose

Enable users to initiate AI-powered outbound calls to individual leads.

Requirements:

- Enter Customer Name.
- Enter Phone Number.
- Select Employee.
- Select Language.
- Add Optional Lead Details.
- Start Call.
- Display Call Status.
- Save Call Result.
- Capture Lead Information.
- Generate Summary.
- Schedule Follow-up.

Call states:

- Queued.
- Initiating.
- Ringing.
- Answered.
- In Progress.
- Completed.
- No Answer.
- Busy.
- Failed.
- Cancelled.

11. MODULE 07 – BULK CAMPAIGNS

11.1 Purpose

Enable users to create and manage outbound AI calling campaigns for multiple contacts.

Requirements:

- Create Campaign.
- Campaign Name.
- Campaign Description.
- Upload Contacts.
- Import Google Sheets.
- Select AI Employee.
- Configure Calling Schedule.
- Configure Retry Rules.
- Configure Priorities.
- Start Campaign.
- Pause Campaign.
- Resume Campaign.
- Stop Campaign.
- Monitor Progress.
- View Campaign Results.
- Export Data.

11.2 Campaign Analytics

- Total Contacts.
- Pending Calls.
- Completed Calls.
- Answered Calls.
- Failed Calls.
- Successful Leads.
- Conversion Rate.

- Credits Used.

Calling must respect applicable consent, DND, DLT, calling-hour restrictions, and telephony provider policies.

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12. MODULE 08 – INBOUND CALLS

12.1 Purpose

Enable organizations to receive customer calls through configured phone numbers and AI employees.

Requirements:

- Configure Inbound Numbers.
- Assign AI Employees.
- Configure Welcome Messages.
- Configure Call Routing.
- Handle Incoming Calls.
- Manage Concurrent Calls.
- Capture Customer Information.
- Generate Call Summary.
- Save Transcript.
- Store Recording Metadata.
- Transfer to Human Agent.

Detailed routing rules will be validated during reverse engineering.

---

13. MODULE 09 – LEADS & RESULTS

13.1 Purpose

Provide centralized lead management.

Requirements:

- View All Leads.
- Search Leads.
- Filter Leads.
- View Lead Details.

- View Associated Calls.
- Manage Lead Status.
- Manage Qualification.
- Assign Leads.
- Schedule Follow-ups.
- Export Leads.

Lead fields:

- Lead ID.
- Name.
- Phone Number.
- Email.
- Source.
- Assigned Employee.
- Status.
- Qualification Status.
- Last Contact Date.
- Follow-up Date.
- Summary.

---

14. MODULE 10 – ALL CONVERSATIONS

14.1 Purpose

Provide centralized conversation history for inbound and outbound calls.

Requirements:

- View Conversations.
- Search Contacts.
- Filter Employee.
- Filter Date.
- Filter Outcome.
- Filter Sentiment.

- View Call Details.
- View Transcript.
- Play Recording.
- View AI Summary.
- View Captured Information.
- View Credits Used.
- Export Data.

Conversation details:

- Contact.
- Employee.
- Call Direction.
- Start Time.
- End Time.
- Duration.
- Call Status.
- Transcript.
- Recording.
- Summary.
- Sentiment.
- Outcome.
- Captured Data.
- Credits Used.

Recording and transcript access must follow organization permissions and retention rules.

---

15. MODULE 11 – TRAIN EMPLOYEES

15.1 Purpose

Provide centralized management of AI employee training and knowledge sources.

Requirements:

- View All Employees.

- Manage Knowledge Sources.
- Upload Documents.
- Add Website URLs.
- Manage FAQs.
- Track Processing Status.
- Update Knowledge.
- Remove Knowledge.
- Test Responses.

The relationship between centralized training and employee-specific training must be validated during reference product analysis.

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16. MODULE 12 – PHONE NUMBERS

16.1 Purpose

Manage telephony numbers used for inbound and outbound calling.

Requirements:

- View Numbers.
- Add / Configure Numbers.
- Assign Numbers to Employees.
- Configure Inbound Routing.
- Configure Outbound Caller ID.
- View Provider.
- View Number Status.
- Manage Assignments.

Number provisioning depends on telephony provider support and applicable Indian regulations.

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17. MODULE 13 – PERFORMANCE

17.1 Purpose

Provide analytics for AI employees, campaigns, calls, and organizations.

Metrics:

- Total Calls.
- Successful Calls.
- Answer Rate.
- Lead Conversion Rate.
- Average Call Duration.
- Average Response Time.
- Employee Performance.
- Campaign Performance.
- Credits Used.
- Cost Per Call.
- Cost Per Qualified Lead.

Requirements:

- Date Filters.
- Comparison Views.
- Charts.
- Employee Reports.
- Campaign Reports.
- Export Reports.

---

18. MODULE 14 – BILLING

18.1 Purpose

Manage subscriptions, credits, payments, and usage.

Requirements:

- Current Subscription.
- Available Credits.
- Credit Usage.
- Call Consumption.
- Subscription Plans.
- Additional Credit Purchase.

- Payment History.
- Invoices.
- Payment Status.
- Razorpay Integration.

Billing must use verified provider usage records and an auditable credit ledger.

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19. MODULE 15 – SETTINGS

19.1 Purpose

Provide organization-level and user-level configuration.

Proposed categories:

- Organization Profile.
- User Profile.
- Team Management.
- Roles and Permissions.
- Language Preferences.
- Notifications.
- Security.
- Calling Preferences.
- Integrations.
- Billing Preferences.

Detailed settings will be finalized through reference product analysis.

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20. MODULE 16 – DEVELOPERS

20.1 Purpose

Provide developers with tools to integrate OWN AI into external systems.

Requirements:

- API Documentation.
- API Key Management.
- Webhook Configuration.

- Webhook Event History.
- API Usage.
- Authentication Documentation.
- Request and Response Examples.
- Integration Guides.
- Error Codes.
- API Rate Limits.

API secrets must be securely generated, stored, and rotated.

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21. INTEGRATION REQUIREMENTS

The platform will support the following integrations.

Integration	Purpose
Supabase	Database, Authentication, Storage
Sarvam AI	Multilingual Voice AI
Sarvam SIP	SIP Telephony Connectivity
Redis	Caching and Queue Management
Celery	Background Processing
n8n	Business Workflow Automation
Google Sheets API	Lead Synchronization
Meta Lead Ads API	Facebook and Instagram Lead Acquisition
WhatsApp Business Cloud API	Messaging and Notifications
Calendly API	Appointment Scheduling
Zoom API	Video Meetings
Razorpay	Payment Processing
Sentry	Error Monitoring
Grafana	Performance Monitoring

All integrations must support secure credential management, validation, error handling, and usage logging.

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22. TECHNICAL ARCHITECTURE

22.1 Architecture Strategy

OWN AI will use a Monorepo architecture with a Modular Monolith backend initially.

The repository will contain:

- Frontend.
- Backend.
- AI Engine.
- Background Workers.
- Automation.
- Database.
- Shared Contracts.
- Infrastructure.
- Tests.
- Documentation.

The architecture must support future separation of high-load services where justified.

22.2 High-Level System Architecture

Frontend:
React + TypeScript + Vite

↓

Backend:
Python + FastAPI

↓

Core Systems:

- Supabase PostgreSQL.
- Supabase Auth.
- Supabase Storage.
- Redis.

- Celery.
- AI Engine.
- Telephony Integration.
- n8n Automation.

External Services:

- Sarvam AI.
- Sarvam SIP.
- WhatsApp Business API.
- Google Sheets.
- Meta Lead Ads.
- Calendly.
- Zoom.
- Razorpay.

Real-time voice streaming and call lifecycle processing must be handled by the backend and AI/telephony layer. n8n should be used for business workflow automation, not as the real-time audio engine.

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23. APPROVED TECHNOLOGY STACK

Layer	Technology	Purpose
Frontend	React + TypeScript	Website and dashboard
UI Styling	Tailwind CSS	Responsive interfaces
UI Components	shadcn/ui	Reusable components
Backend	Python + FastAPI	APIs and business logic
Workflow Automation	n8n	Business workflows
Database	PostgreSQL (Supabase)	Application data
ORM	SQLAlchemy	Database access
Database Migrations	Alembic	Schema versioning
Voice AI	Sarvam AI	Voice conversations

Telephony / SIP	Sarvam AI SIP	SIP calling
Queue Management	Redis	Queue and cache
Background Processing	Celery	Asynchronous jobs
Authentication	Supabase Auth	Authentication
Authorization	RBAC + RLS	Access control
Lead Integration	Google Sheets API	Lead synchronization
Advertisement Integration	Meta Lead Ads API	Lead acquisition
Messaging	WhatsApp Business Cloud API	Messaging
Appointment Scheduling	Calendly API	Appointments
Video Meetings	Zoom API	Meetings
Payments	Razorpay	Billing
Deployment	Docker	Containerization
Reverse Proxy	Nginx	Traffic routing
Error Monitoring	Sentry	Error tracking
Performance Monitoring	Grafana	Operational monitoring
Browser Testing	Playwright	E2E tests
API Testing	Postman / Automated Tests	API verification
Version Control	Git + GitHub	Source control
CI/CD	GitHub Actions	Automated checks

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24. PROGRAMMING ARCHITECTURE

24.1 Frontend Architecture

Technology:

- React.
- TypeScript.
- Vite.

- Tailwind CSS.
- shadcn/ui.

Architecture:
Feature-oriented frontend structure.

Responsibilities:

- Dashboard rendering.
- Navigation.
- Employee configuration.
- Calling interfaces.
- Analytics.
- API integration.
- Form validation.
- Authentication state.
- Loading and error states.

24.2 Backend Architecture

Technology:

- Python.
- FastAPI.
- SQLAlchemy.
- Pydantic.
- Alembic.

Responsibilities:

- Authentication validation.
- Organization management.
- Employee management.
- Campaign management.
- Lead management.
- Call lifecycle.
- Conversation management.

- Training management.
- Billing.
- Integration management.
- Webhook processing.
- Analytics.

All API routes must be versioned under /api/v1.

24.3 AI Engine Architecture

Responsibilities:

- AI Employee Orchestration.
- Prompt Management.
- Speech-to-Text.
- LLM Processing.
- Text-to-Speech.
- Conversation State.
- Knowledge Retrieval.
- Language Configuration.
- Provider Integration.

The system should use provider adapters so that additional AI providers can be added without changing core business logic.

24.4 Background Processing

Technology:
Redis + Celery.

Responsibilities:

- Campaign Execution.
- Scheduled Calls.
- Retry Handling.
- Follow-up Scheduling.
- Lead Processing.
- Notifications.

- Report Generation.
- Analytics Aggregation.

24.5 Workflow Automation

Technology:
n8n.

Responsibilities:

- Lead Processing.
- CRM Synchronization.
- Appointment Workflows.
- Follow-up Automation.
- WhatsApp Notifications.
- External Webhooks.

---

25. COMPLETE PROGRAMMING FOLDER STRUCTURE

Repository Name: AOTMS_AI_2027

```
AOTMS_AI_2027/
|
├── frontend/
|   ├── public/
|   ├── src/
|   |   ├── assets/
|   |   ├── components/
|   |   |   ├── ui/
|   |   |   ├── layout/
|   |   |   ├── charts/
|   |   |   └── common/
|   |   ├── pages/
|   |   |   ├── dashboard/
|   |   |   ├── employees/
```

```
|   |   |   ├── calling/
|   |   |   ├── leads/
|   |   |   ├── conversations/
|   |   |   ├── training/
|   |   |   ├── phone-numbers/
|   |   |   ├── performance/
|   |   |   ├── billing/
|   |   |   ├── settings/
|   |   |   └── developers/
|   |   ├── features/
|   |   |   ├── auth/
|   |   |   ├── employees/
|   |   |   ├── campaigns/
|   |   |   ├── instant-leads/
|   |   |   ├── inbound-calls/
|   |   |   ├── conversations/
|   |   |   └── integrations/
|   |   ├── hooks/
|   |   ├── services/
|   |   ├── stores/
|   |   ├── types/
|   |   ├── utils/
|   |   ├── routes/
|   |   ├── App.tsx
|   |   └── main.tsx
|   ├── package.json
|   └── vite.config.ts
|
```

```
├── backend/
│   ├── app/
│   │   ├── main.py
│   │   ├── core/
│   │   │   ├── config.py
│   │   │   ├── security.py
│   │   │   ├── logging.py
│   │   │   └── exceptions.py
│   │   ├── api/
│   │   │   └── v1/
│   │   │       ├── auth.py
│   │   │       ├── organizations.py
│   │   │       ├── employees.py
│   │   │       ├── campaigns.py
│   │   │       ├── leads.py
│   │   │       ├── calls.py
│   │   │       ├── conversations.py
│   │   │       ├── training.py
│   │   │       ├── phone_numbers.py
│   │   │       ├── billing.py
│   │   │       ├── integrations.py
│   │   │       └── webhooks.py
│   │   ├── modules/
│   │   │   ├── auth/
│   │   │   ├── organizations/
│   │   │   ├── employees/
│   │   │   ├── campaigns/
│   │   │   ├── leads/
```

```
|   |   |   ├── calls/
|   |   |   ├── conversations/
|   |   |   ├── training/
|   |   |   ├── phone_numbers/
|   |   |   ├── billing/
|   |   |   └── analytics/
|   |   ├── integrations/
|   |   |   ├── supabase/
|   |   |   ├── sarvam/
|   |   |   ├── telephony/
|   |   |   ├── n8n/
|   |   |   ├── whatsapp/
|   |   |   ├── google_sheets/
|   |   |   ├── meta_ads/
|   |   |   ├── calendly/
|   |   |   ├── zoom/
|   |   |   └── razorpay/
|   |   ├── database/
|   |   ├── middleware/
|   |   ├── schemas/
|   |   └── utils/
|   ├── tests/
|   ├── requirements.txt
|   └── pyproject.toml
|
├── ai-engine/
|   ├── agents/
|   ├── prompts/
```

```
|   ├── models/
|   ├── speech_to_text/
|   ├── text_to_speech/
|   ├── conversation/
|   ├── orchestration/
|   ├── knowledge_base/
|   └── providers/
|
├── workers/
|   ├── celery_app.py
|   ├── tasks/
|   |   ├── calling/
|   |   ├── campaigns/
|   |   ├── leads/
|   |   ├── followups/
|   |   ├── notifications/
|   |   └── reports/
|   └── schedules/
|
├── automation/
|   ├── workflows/
|   |   ├── lead-processing/
|   |   ├── follow-ups/
|   |   ├── appointments/
|   |   ├── crm-sync/
|   |   └── notifications/
|   └── README.md
|
```

```
├── database/
|   ├── migrations/
|   ├── seeds/
|   ├── functions/
|   ├── policies/
|   ├── schemas/
|   └── database-diagrams/
|
├── shared/
|   ├── api-contracts/
|   ├── event-schemas/
|   └── constants/
|
├── infra/
|   ├── docker/
|   ├── nginx/
|   ├── monitoring/
|   ├── grafana/
|   └── deployment/
|
├── tests/
|   ├── unit/
|   ├── integration/
|   ├── e2e/
|   ├── security/
|   └── load/
|
├── docs/
```

```
|   ├── PRD/
|   ├── REVERSE_ENGINEERING/
|   ├── ARCHITECTURE/
|   ├── DATABASE/
|   ├── API/
|   ├── WORKFLOWS/
|   └── DEPLOYMENT/
|
├── scripts/
|   ├── setup/
|   ├── development/
|   ├── database/
|   └── deployment/
|
├── .github/
|   └── workflows/
|
├── .env.example
├── .gitignore
├── docker-compose.yml
├── README.md
└── LICENSE
```

25.1 Folder Responsibilities

Directory	Responsibility
frontend	Complete web application
backend	APIs and business logic
ai-engine	AI voice processing

workers	Background processing
automation	n8n workflows
database	PostgreSQL schema and migrations
shared	Shared contracts and constants
infra	Deployment and monitoring
tests	Automated testing
docs	Product and technical documentation
scripts	Setup and deployment scripts

25.2 Implementation Rule

The folder structure is a target architecture.

The development team must inspect the existing repository before creating files. Do not overwrite existing working code.

Create files progressively as their modules are approved.

Do not create unnecessary empty files or deploy every component as a separate microservice during the initial phase.

---

26. DATABASE ARCHITECTURE

Primary Database: Supabase PostgreSQL.

ORM: SQLAlchemy.

Migration Tool: Alembic.

26.1 Proposed Database Domains

Organization and Access

- organizations
- users
- organization_members
- roles
- permissions
- role_permissions

AI Employees

- employees
- employee_languages
- employee_prompts
- employee_voices
- employee_models
- employee_settings
- employee_prompt_versions

Employee Training

- training_sources
- training_documents
- training_jobs
- knowledge_chunks
- employee_knowledge_sources

Leads and Contacts

- contacts
- leads
- lead_sources
- lead_assignments
- lead_activities
- followups

Campaigns

- campaigns
- campaign_contacts
- campaign_schedules
- campaign_attempts

Calls and Conversations

- calls
- conversations

- transcripts
- recordings
- conversation_summaries
- conversation_outcomes
- extracted_information

Calling Infrastructure

- phone_numbers
- phone_number_assignments
- telephony_providers
- call_events

Actions and Automation

- employee_actions
- action_executions
- workflows
- workflow_executions
- webhooks

Billing

- subscription_plans
- subscriptions
- credit_accounts
- credit_ledger
- usage_records
- payments
- invoices

Integrations

- integrations
- integration_credentials
- integration_events
- integration_logs

Monitoring and Security

- audit_logs
- api_keys
- notification_logs

These are proposed logical entities. Exact table definitions, foreign keys, indexes, constraints, and relationships must be designed during database engineering.

26.2 Database Requirements

- Organization-level data isolation.
- PostgreSQL Row Level Security.
- Foreign key constraints.
- Appropriate indexes.
- Transaction management.
- Database migrations.
- Backup and recovery.
- Audit trails.
- Usage ledger integrity.

---

27. API ARCHITECTURE

Backend: FastAPI.

Base Route:

/api/v1

27.1 Proposed API Groups

API Group	Purpose
/auth	Authentication validation
/organizations	Organization management
/users	User management
/employees	AI employee operations
/campaigns	Campaign operations

/leads	Lead management
/calls	Calling operations
/conversations	Conversation management
/training	Employee training
/phone-numbers	Number management
/performance	Analytics
/billing	Subscription and usage
/integrations	External integrations
/webhooks	Provider webhooks
/developers	API and developer tools

27.2 API Standards

- Versioned APIs.
- Request validation using Pydantic.
- Consistent response schemas.
- Standardized error responses.
- Authentication and authorization.
- Rate limiting.
- Pagination.
- Filtering and sorting.
- Idempotency for critical operations.
- API documentation using OpenAPI.

---

28. AI VOICE PROCESSING ARCHITECTURE

28.1 Core Processing Components

- Speech-to-Text.
- LLM Processing.
- Text-to-Speech.

- Conversation State.
- Prompt Management.
- Voice Configuration.
- Language Detection where supported.
- Knowledge Retrieval.
- Call State Management.

28.2 Provider Architecture

Initial Voice AI Provider: Sarvam AI.

The provider integration must be isolated behind internal interfaces.

Future providers may be integrated without rewriting the employee management and campaign modules.

28.3 Real-Time Calling Requirements

- Low-latency audio processing.
- Call state tracking.
- Audio stream management.
- Conversation turn management.
- Error recovery.
- Call termination handling.
- Provider event handling.
- Call recording metadata.

Actual latency and concurrency targets must be finalized through provider testing.

---

29. WORKFLOW AUTOMATION ARCHITECTURE

Automation Platform: n8n.

29.1 Initial Workflows

Lead Processing

New Lead → Validate → Save Lead → Trigger Calling Job.

Follow-up

Call Completed → Evaluate Outcome → Schedule Follow-up → Notify User.

Appointment

Customer Requests Appointment → Check Availability → Create Booking → Send Confirmation.

CRM Synchronization

Lead Captured → Validate Data → Update CRM → Save Integration Result.

WhatsApp Notification

Call Completed → Prepare Message → Send Notification → Log Result.

29.2 Workflow Requirements

- Workflow versioning.
- Execution history.
- Error handling.
- Retry policies.
- Credential security.
- Webhook authentication.
- Execution monitoring.

Real-time call processing must not depend on ordinary n8n execution.

---

30. SECURITY REQUIREMENTS

- Secure authentication.
- Supabase Auth integration.
- Role-Based Access Control.
- Organization-level tenant isolation.
- PostgreSQL Row Level Security.
- Secure credential storage.
- Password hashing where custom credentials exist.
- JWT validation.
- Refresh token rotation and revocation.
- API rate limiting.
- Input validation.
- Webhook signature verification.

- Audit logging.
- Recording and transcript access control.
- Backup and recovery.
- No secrets in source code.
- No API keys committed to GitHub.

Production credentials must be stored in deployment secrets or an approved secrets manager.

---

31. NON-FUNCTIONAL REQUIREMENTS

Performance

- Responsive dashboard.
- Efficient API responses.
- Optimized database queries.
- Asynchronous job processing.
- Efficient call data retrieval.

Reliability

- Retry handling.
- Idempotent webhook processing.
- Call state reconciliation.
- Background job monitoring.
- Error logging.
- Operational alerts.

Scalability

- Multi-tenant architecture.
- Scalable worker processes.
- Redis queue management.
- Independent scaling of high-load components when necessary.

Usability

- Clear navigation.
- Guided employee creation.

- Search and filtering.
- Responsive design.
- Clear call status.
- Accessible interface components.
- Consistent UI design system.

Maintainability

- Modular source code.
- Type annotations.
- API documentation.
- Database migrations.
- Automated tests.
- Code review.
- CI/CD checks.

---

32. TESTING REQUIREMENTS

Unit Testing

Test individual modules and business logic.

Integration Testing

Test:

- Supabase connectivity.
- FastAPI and database.
- Redis and Celery.
- AI provider integrations.
- Telephony integrations.
- n8n webhooks.
- Razorpay integration.

Frontend Testing

Test:

- Navigation.

- Forms.
- Employee creation.
- Campaign configuration.
- Dashboard interactions.

End-to-End Testing

Test complete workflows:

1. User Registration.
2. Organization Creation.
3. Employee Creation.
4. Employee Training.
5. Test Conversation.
6. Instant Lead Calling.
7. Bulk Campaign.
8. Inbound Call.
9. Conversation Review.
10. Billing and Usage.

Security Testing

- Authentication.
- Authorization.
- Tenant Isolation.
- RLS.
- API Security.
- Secrets Exposure.
- Webhook Security.

Load Testing

Evaluate:

- Concurrent calls.
- Campaign job processing.
- API throughput.

- Database load.
- Worker capacity.

---

33. DEPLOYMENT AND INFRASTRUCTURE

33.1 Deployment Technology

- Docker.
- Nginx.
- Supabase Cloud.
- Cloud Server.
- GitHub Actions.
- Sentry.
- Grafana.

33.2 Environments

Development

Local development and testing.

Staging

Pre-production validation and integration testing.

Production

Live customer application.

33.3 Deployment Requirements

- Containerized services.
- Secure environment configuration.
- Database migrations.
- Health checks.
- Centralized logs.
- Error monitoring.
- Backup strategy.
- Rollback strategy.
- CI/CD validation.

34. DEVELOPMENT ROADMAP

Phase 1 – Product Discovery

- Analyze existing PRD.
- Reverse engineer reference product.
- Document modules.
- Document workflows.
- Identify gaps.
- Finalize functional requirements.

Phase 2 – Architecture

- Finalize system architecture.
- Finalize folder structure.
- Design database schema.
- Design API specifications.
- Define security model.
- Define integration contracts.

Phase 3 – Foundation

- GitHub repository setup.
- Frontend project setup.
- Backend project setup.
- Supabase configuration.
- Redis configuration.
- Docker development environment.
- Authentication.
- Organization and tenant management.

Phase 4 – Frontend Development

- Dashboard.
- Employee Listing.
- Employee Creation.

- Employee Workspace.
- Calling Pages.
- Leads.
- Conversations.
- Training.
- Performance.
- Billing.
- Settings.

Phase 5 – Backend Development

- Employee APIs.
- Lead APIs.
- Campaign APIs.
- Calling APIs.
- Conversation APIs.
- Training APIs.
- Billing APIs.
- Integration APIs.

Phase 6 – AI Voice Engine

- Sarvam integration.
- STT.
- LLM orchestration.
- TTS.
- Prompt configuration.
- Voice selection.
- Conversation state management.

Phase 7 – Telephony

- SIP connectivity.
- Outbound calls.
- Inbound calls.

- Call state handling.
- Webhooks.
- Call recording metadata.

Phase 8 – Automation

- n8n workflows.
- Lead automation.
- Follow-ups.
- CRM integrations.
- Appointment workflows.
- WhatsApp notifications.

Phase 9 – Billing and Analytics

- Usage tracking.
- Credit ledger.
- Razorpay.
- Dashboard analytics.
- Campaign reports.
- Employee performance.

Phase 10 – Production Readiness

- End-to-end testing.
- Security validation.
- Load testing.
- Monitoring.
- Deployment.
- Documentation.
- Production release.

---

35. DEVELOPMENT TIME ESTIMATION

Assumption: Two full-time developers using Cursor AI, with requirements finalized and API access available.

Layer	Technology	Estimated Time
Frontend	React + TypeScript	2–3 weeks
UI Styling	Tailwind CSS	3–5 days
Backend	Python + FastAPI	2–3 weeks
Workflow Automation	n8n	1–2 weeks
Database	Supabase PostgreSQL	3–5 days
ORM	SQLAlchemy	2–3 days
Voice AI	Sarvam AI	1–2 weeks
Telephony / SIP	Sarvam AI SIP	4–7 days
Queue Management	Redis	2–3 days
Background Processing	Celery	3–5 days
Authentication	Supabase Auth + JWT validation	2–3 days
Lead Integration	Google Sheets API	2–3 days
Advertisement Integration	Meta Lead Ads API	3–5 days
Messaging	WhatsApp Business Cloud API	3–5 days
Appointment Scheduling	Calendly API	2–3 days
Video Meetings	Zoom API	2–3 days
Deployment	Docker	3–5 days
Error Monitoring	Sentry	1–2 days
Performance Monitoring	Grafana	2–3 days

Estimation Notes

- These are initial engineering estimates.
- Estimates overlap and should not be added directly to calculate calendar duration.
- Product discovery, telephony testing, security hardening, and production QA may increase the total timeline.
- External API approvals and provider limitations may affect delivery.

- Two developers must coordinate frontend, backend, and integration tasks.

---

36. MVP RELEASE SCOPE

The first release should prioritize:

1. User Registration and Login.
2. Organization Setup.
3. Dashboard.
4. AI Employee Creation.
5. Custom Prompt Configuration.
6. Language and Voice Selection.
7. Employee Training.
8. Talk to an Employee.
9. Instant Leads.
10. Basic Bulk Campaigns.
11. Inbound Calling.
12. Call History.
13. Conversation Transcripts.
14. Lead Management.
15. Basic Performance Analytics.
16. Usage Tracking.
17. Basic Billing.

Advanced CRM marketplace, enterprise reporting, and advanced AI analytics can follow after MVP validation.

---

37. ACCEPTANCE CRITERIA

The MVP will be considered ready when:

- Users can register and create organizations.
- Users can create AI employees.
- Users can configure prompts, languages, and voices.

- Users can test employees.
- Users can initiate authorized outbound calls.
- Users can receive configured inbound calls.
- Call results are stored accurately.
- Conversations and transcripts can be reviewed.
- Leads can be captured and managed.
- Usage and credits are tracked.
- Organization data remains isolated.
- Critical workflows pass automated tests.
- Production monitoring and error handling are operational.

---

38. REVERSE ENGINEERING VALIDATION

Outpero AI is the reference product for authorized feature and workflow observation.

The reverse engineering process must document:

- Screens and navigation.
- User workflows.
- Forms and inputs.
- Actions and outcomes.
- Validation rules.
- Loading and error states.
- Available integrations.
- User roles.
- Calling behavior.
- Reporting behavior.

38.1 Requirement Classification

Each requirement must be classified as:

Observed: Directly verified in the reference product.

Proposed: Recommended functionality for OWN AI.

Pending Validation: Requires further inspection or clarification.

38.2 Validation Status

Module	Current Status
Dashboard	Initial requirements documented
Employees	Initial requirements documented
New Employee	Initial requirements documented
Employee Workspace	Initial requirements documented
Talk to an Employee	Proposed requirements
Instant Leads	Initial requirements documented
Bulk Campaigns	Initial requirements documented
Inbound Calls	Initial requirements documented
Leads & Results	Proposed requirements
All Conversations	Proposed requirements
Train Employees	Proposed requirements
Phone Numbers	Proposed requirements
Performance	Proposed requirements
Billing	Proposed requirements
Settings	Proposed requirements
Developers	Proposed requirements

The initial PRD is not a verified specification of all reference product functionality.

Each module must be compared with actual product behavior before its final implementation requirements are approved.

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39. FUTURE SCOPE

Potential future capabilities:

- Advanced AI Agent Builder.
- Visual Conversation Flow Builder.
- Integrations Marketplace.

- Advanced CRM Marketplace.
- Custom Voice Configuration.
- Advanced Knowledge Base.
- Advanced RAG.
- Real-Time Call Monitoring.
- Human Agent Handoff.
- Team Collaboration.
- White-Label SaaS.
- Enterprise Analytics.
- Advanced Usage-Based Billing.
- Additional AI and telephony providers.
- More Indian and international languages.
- Advanced Quality Assurance.
- Automated Conversation Evaluation.

These features require separate prioritization and approval.

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40. FINAL PROJECT PRINCIPLES

1. Existing PRD and approved requirements are the source of truth.
2. Outpero AI is a reference for product discovery, not a source of proprietary code.
3. No module should be implemented without its approved requirements.
4. The original PRD must remain unchanged as a baseline.
5. New observations must be documented before modifying the final specification.
6. Supabase PostgreSQL is the primary database.
7. FastAPI is responsible for core backend business logic.
8. n8n handles business workflow automation.
9. Redis and Celery handle background processing.
10. AI and telephony integrations must use modular provider adapters.
11. Security and tenant isolation must be implemented from the beginning.
12. Development must proceed incrementally with testing and review.

13. Existing working files must never be overwritten without inspection and approval.
14. Production credentials must never be committed to GitHub.

DOCUMENT CONTROL

Product: OWN AI
Organization: Academy of Tech Masters
Reference Product: Outpero AI
Document: Complete Product Requirements Document
Version: 1.0
Status: Consolidated Initial Draft
Next Milestone: Complete Reverse Engineering Audit and Final PRD Approval

End of Document.